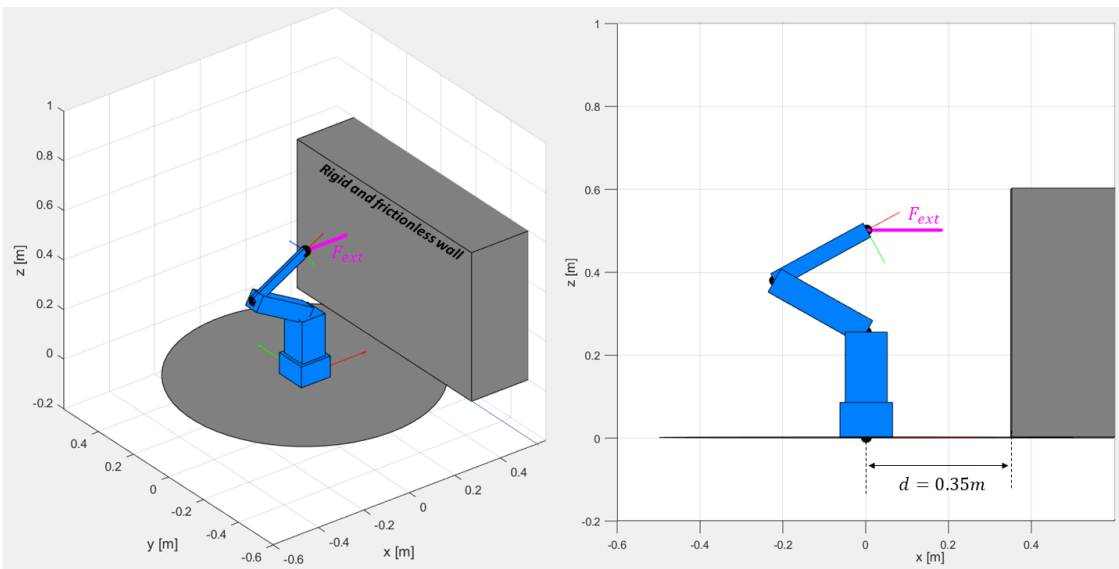
ME446/SE422/ECE489 Robot Dynamics and Control

Spring 2026 Justin Yim **Homework 9**

Homework is due on Gradescope by 11:59 pm. In your homework submission, clearly demonstrate how you obtained your final answer including all derivations and any MATLAB code you wrote. Include the names of anyone you worked with and cite any resources, including AI tools, you used.

1. Contact and Impact

In this problem, you will implement contact models for a robot manipulator end effector interacting with walls in the environment.



(a) Modify the `wall_contact.m` function to return the distance from the wall to the end effector in units of meters. This value should be positive when the end effector is not in contact with the wall and be negative when the end effector is through the wall. When this value becomes zero, the robot transitions from free to contact mode. Modify the `fcn_Jwall.m` function to return the Jacobian of `wall_contact.m` with respect to the joint coordinates. Modify the `fcn_dJwall.m` function to return the time derivative of the Jacobian. (HINT: `fcn_p4.m` solves FK for the end effector position, `fcn_J4.m` returns its Jacobian, and `fcn_dJ4.m` returns the Jacobian time derivative). Report your code. (15 points)

(b) Take a look at the unconstrained forward dynamics in `UNCONS_dyn_manip.m` used when the robot is in free space. Modify the `CONS_dyn_manip.m` function to compute the constrained forward dynamics of the robot when the end effector is in contact with the wall using the functions you wrote in the previous part. `CONS_dyn_manip.m` should also return the constraint force F_c that the wall exerts on the robot. When this value becomes zero, the robot transitions from contact mode to free mode. Remember that F_c is the force the wall exerts on the robot along the direction normal to the wall surface and it should be positive as the wall pushes the robot away. Report your code. (15 points)

(c) Modify the `impact_reset.m` function to return the new initial condition of the robot in contact

mode after it impacts the surface using the impact law. From $q = [0; -0.8108; 1.6215]$ and $\dot{q}^- = [0; 1; -1]$, what is $\dot{q}^+$? Report your answer for $\dot{q}^+$ and your code. (10 points)

(d) Implement a force controller:

$$\tau = J(q)^T \lambda_d + \hat{C}(q, \dot{q})\dot{q} + \hat{G}(q)$$

and command the robot to apply a force of 10 N in the positive x_0 direction and 0 N in the y_0 and z_0 directions. What is the sign of F_c in this case? Does it match your intuition? Change the initial joint angles to `q0 = [0;-1.4;1.7]` and run the simulation. Does the robot impact the wall and settle into applying 10 N directly into the wall? Report the normal force exerted by the wall (figure 5) and your code. (10 points)

(e) Set the initial joint angles back to `q0 = [0; -2.63; 2.13]`. Modify fcn_trajectory.m to follow the task-space trajectory (don't forget to compute the time derivatives as well):

$$X_d = \begin{bmatrix} 0.35 \\ 0.1sin(2t) \\ 0.45 + 0.05cos(2t) \end{bmatrix}$$

Implement a hybrid force-position controller (with gravity and Coriolis compensation) that draws this ellipse on the wall while applying a force of 10 N into the wall in the x_0 direction. Can you get the task-space y and z error to be less than 5 mm while the robot is touching the wall (it's ok if the error is greater before contact)? Report:

- your code
- your K_p and K_d
- The joint, velocity, and torque trajectories over time (figure 1)
- The plot of task-space error (figure 2)
- The plot of wall normal force (figure 5)

(30 points)

(f) Write a soft contact force model into function `fcn_ExternalForce.m` for a surface that is 0.3 m away from the robot (rather than 0.35 m away). This soft contact model will involve two steps:

i. Write a conditional statement for when the contact force is applied

ii. Write a spring-damper force that pushes the end effector out of the surface and assume no friction.

Note that `fcn_ExternalForce.m` specifies force vectors applied to the end effector with respect to frame 0, NOT with respect to the surface normal as in the above problems. Do the signs of your forces match or differ due to this change?

Use the same trajectory and controller. Tune your soft contact model spring constant and damping constant until the robot arm progresses no further than 1 cm past 0.3 m (that is, it gets no closer than 4 cm to the rigid wall) and there is little oscillation. Report:

- your code
- your chose stiffness and damping
- The joint, velocity, and torque trajectories over time (figure 1)
- The plot of distance to the wall (figure 3)
- The plot of soft force (figure 5)

(20 points)

Approximately how many hours did it take to complete this homework assignment?